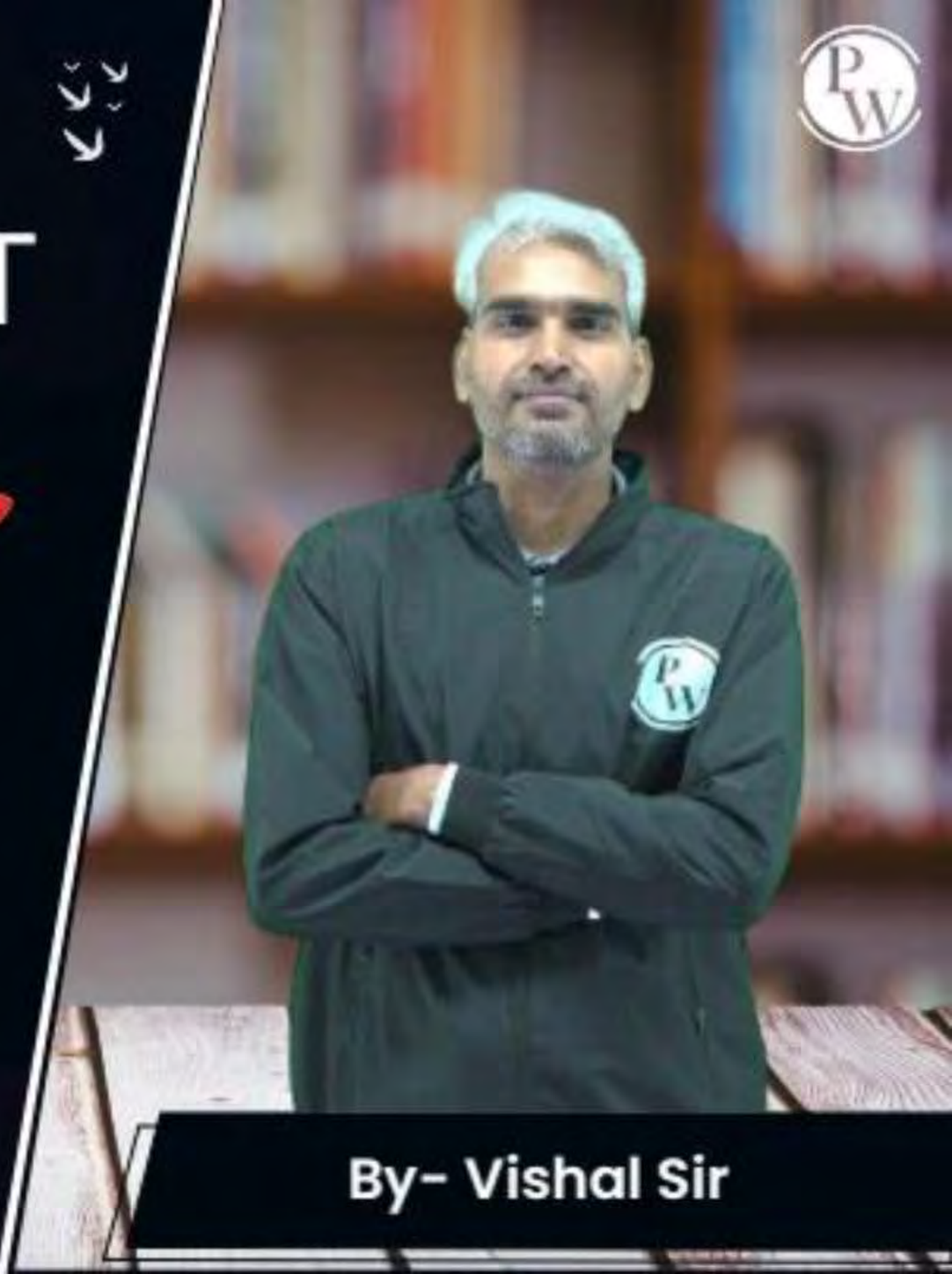
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



Topic — Introduction to Discrete Maths

Topic — Set

Topic — Representation of Set

Topic — Types of sets

Slide

Topics to be Covered



Topic Types of sets

Topic Terminologies related to sets

Slide



Topic : Set

- A well-defined unordered collection of distinct elements is called as set.

$A = \{1, 2, 3, 4, Jan, Mon\} = \{Jan, 1, 2, 3, 4, Mon\}$

$B = \{1, 2, 3, 4, 5\}$

$C = \{a, e, i, o, u\}$

Slide



Topic : Representation of sets

- There are three different ways in which set may be represented.

1. Roaster form or Tabular form:
2. Set-builder form:
3. Statement form:



Topic : Roaster form or Tabular form

In this notation we will list all the elements of the set within the curly braces.

eg: $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, Jan, feb\}$

{We may not represent the elements of set B using "set-builder" notation}

$X = \{1, 2, 3, 4, \ldots\ldots\}$ = Set of all Natural Numbers

$D = \{0, 0.25, 0.3, 0.31, 0.4, 0.79, 0.8, 0.9\}$

Slide



Topic : Set-builder form

* In this notation we define the property and all the elements satisfying that property will be members of the set

$$A = \{ x \mid x \in N \text{ and } x \leq 5 \} = \{1, 2, 3, 4, 5\}$$

- A : it is a set
- x : containing elements x
- $\mid$: such that
- $x \in N$ and $x \leq 5$: Property

$$C = \{ x : x \in R \text{ and } 0 \leq x \leq 1 \}$$

- x : such that
- R : Real number

We can not represent set C in roaster form

Slide



Topic : Statement form

A = Set of all natural numbers less than or equal to 5.

C = Set of all real numbers between 0 & 1

Slide



Topic : Cardinality of a set

{ Cardinality is generally defined for finite sets }

Cardinality of a set A is denoted by $|A|$ and $|A|$ represent the number of elements in set A.

eg, $A = \{1, 2, 3, Jan, Feb\}$

$|A| = 5$

Slide



Topic : Types of sets

Empty Set: A set with no element in it is called an Empty set.

Empty set can be denoted by $\{\ \}$ or $\emptyset$

$$\{\ \} = \emptyset$$

$$|\{\ \}| = |\emptyset| = 0$$

$$|\{1, \{2,3\}\}| = 2$$

Slide



Topic : Types of sets

Singleton Set: A set with exactly '1' element in it is called a Singleton set

eg $A = \{a\}$

$\{1\}$

$\{\{2, \{3, 4\}\}\}$

$\{\{1, \{1, 2\}\}\}$

$\{\{\}\}$

its a set

Containing one element & that element is an Empty set

Slide

$\{\} = \emptyset$

$\therefore \quad |\{\}| = |\emptyset| = 0$

$\{\}$: it is an Empty set

$\{\emptyset\} = \{\{\}\}$: It is a set Containing one element that Element is an Empty set

$|\{\}| = 0$ but $|\{\{\}\}| = 1$

$$\{\} \neq \{\{\}\}$$



Topic : Types of sets

Finite Set: A set that contains finite number of elements is called a finite set

eg: $\{1\}$, $\{1,2\}$, $\{a,b,c\}$ etc..

$\{\ \}$ ← Empty set is always a finite set.

Slide



Topic : Types of sets

Infinite Set: A set that contains infinite number of elements in it is called an infinite set.

Eg:
- A = Set of all natural numbers
- B = Set of all real numbers
- C = Set of all Even integers

Slide

Another Classification w.r.t. number of elements in the set

- Finite sets
  - Finite sets are always Countable
- Infinite sets
  - Countably infinite sets
  - Uncountably infinite sets

A = Set of all natural numbers.

A = { 1, 2, 3, - - - - - . (5000)? , - - - (900000)? - - - - }

A is a Countably infinite set.

B = Set of all non-negative real numbers.

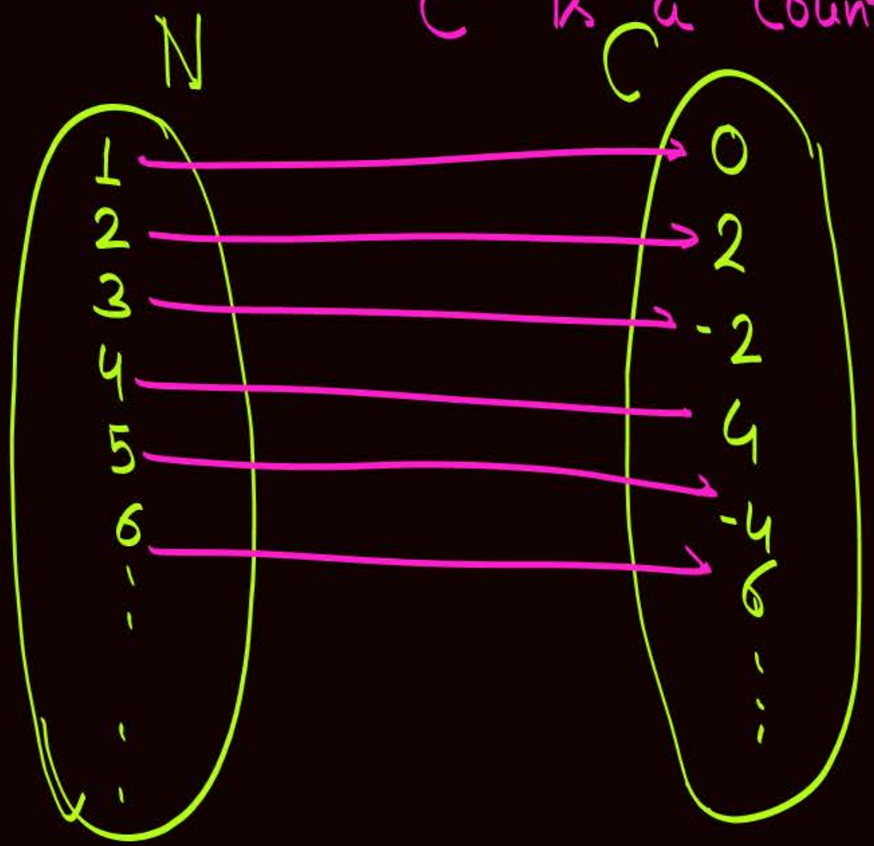
B = { 0, - - - }

B is a Uncountably infinite set

C = Set of all Even integers.

C = { 0, 2, -2, 4, -4, - - - . . . (2000)? . . . (-5000)? . . . }

'C' is a Countably infinite set



$f : N \rightarrow C$

$$f(x) = \begin{cases} 0 & \text{if } x = 1 \\ x & \text{if } x = \text{Even} \\ 1-x & \text{if } x = \text{odd, but } x \neq 1 \end{cases}$$

Let $f:$ is a function from set A to set B

then A = Domain of function f.

B = Co-domain of function f.

One-one: If distinct element of the domain have distinct images in Co-domain then function is one-one

i.e: if $a \neq b$, then $f(a) \neq f(b)$

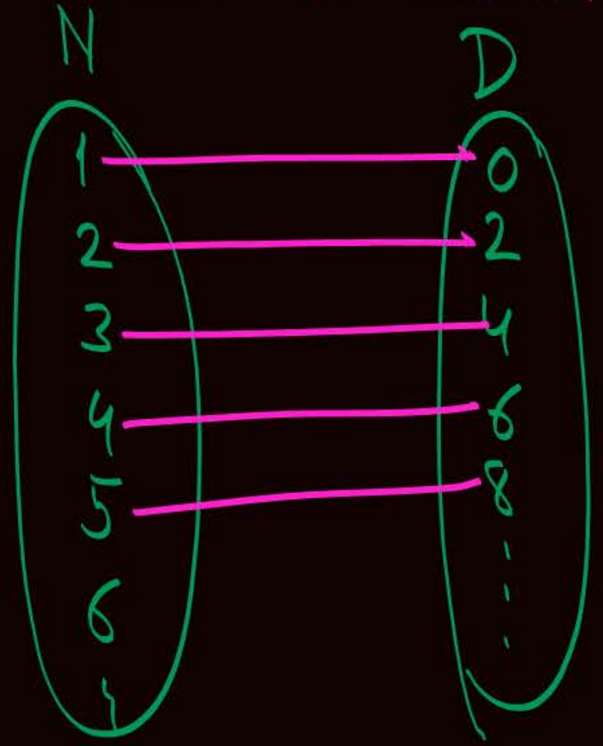
Onto: A function from set A to set B is called onto iff every element of set B (Codomain) is mapped by at least one element of set A (i.e. domain).

i.e. Range of function f = Codomain of function f

D : Set of non-negative Even integers

$D = \{0, 2, 4, 6, \ldots\ldots, (2000)?\ldots\ldots\}$

D is a Countably infinite set.



$$f(x) = 2x - 2$$



Topic : Types of sets

Equal Sets: To sets A & B are said to be equal if and only if every element of set A is a member of set B and every element of set B is a member of set A.

eg: $A = \{1, 2, a, b, c\}$ $B = \{1, b, a, 2, c\}$

Here $A = B$

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eg $A = \{1, 2, 3, 4\}$

$B = \{1, 2, 3, a\}$

$\Longrightarrow A \neq B$

eg: $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, 5\}$

$\Longrightarrow A \neq B$



Topic : Types of sets

Equivalent Sets: Two finite sets are said to be Equivalent if and only if $|A| = |B|$

i.e. same cardinalities

Slide

eg: $A = \{1, 2, a, b, C\}$

$B = \{1, a, b, 2, C\}$

$\Rightarrow A = B$ & $|A| = |B|$ (Equivalent)

$\therefore A \simeq B$

eg: $A = \{1, 2, 3, 4\}$

$B = \{1, 2, 3, a\}$

$\Rightarrow A \neq B$, but $|A| = |B|$

$\therefore A \simeq B$

eg: $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, 5\}$

$\Rightarrow A \neq B$, and $|A| \neq |B|$

$\therefore A \not\simeq B$

Note: ① If two sets are equal, then they are Equivalent as well

② If two sets are equivalent, then they may or may not be equal.

③ If two sets are not Equivalent, then they can never be equal

Note: General definition for two sets to be called Equivalent

- Two sets A & B are called Equivalent sets if we can define bijective (one-one as well as onto) function b/w A & B.

⇓

Two infinite sets A & B can also be Equivalent iff we can define a bijective function b/w A & B.

We can define bijective (one-one + onto) functions b/w set A & set B only if

$$|A| = |B|$$

For one-one & for onto

$|A| \leq |B|$ & $|A| \geq |B|$

eg. let A = Set of all natural nos.

i.e. $A = \{1, 2, 3, 4, \ldots\}$

& B = Set of all positive Even integers

i.e. $B = \{2, 4, 6, 8, \ldots\}$

A & B are Equivalent

We can define a bijective function $f: A \to B$

s.t. $f(x) = 2x$

eg: Let A = Set of all real nos b/w 0 & 1
& B = Set of all natural nos.

↳ We can not define any bijective function b/w A & B.

∴ A & B are not Equivalent sets.



Topic : Types of sets

Universal Set:- It is a set of all elements associated with problem Under Consideration.

Generally denoted by U

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Topic : Types of sets

Subset & Superset:

A subset B → $A \subseteq B$

A superset B → $A \supseteq B$

Subset

Superset

- A is a subset of B if and only if B is a superset of A

{ let A and B are any two sets, if Every element of set A is present in set B then we say that A is a subset of B (or B is a superset A)

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eg: $A = \{1, 2, a, b, c\}$
$B = \{1, a, b, 2, c\}$

Every element of set A is present in B $\therefore A \subseteq B$
Every element of set B is present in A $\therefore B \subseteq A$

if $A \subseteq B$ & $B \subseteq A$ then $A = B$

eg $A = \{1, 2, 3, 4\}$
$B = \{1, 2, 3, a\}$

$A \nsubseteq B$ & $B \nsubseteq A$

if Either $A \nsubseteq B$ or $B \nsubseteq A$ then $A \neq B$

eg: $A = \{1, 2, 3, 5\}$
$B = \{1, 2, 3, 4, 5\}$

$A \subseteq B$ but $B \nsubseteq A$

if Either $A \nsubseteq B$ or $B \nsubseteq A$ then $A \neq B$

① $\emptyset$ is a subset of Every set.

i.e wrt. any set A,

$\emptyset \subseteq A$ is always true

② Every set is a subset of it self.

i.e; wrt. any set A

$A \subseteq A$ is always true



Topic : Types of sets

Proper Subset: For a set A, any subset of set A Except set A itself will be proper subset of set A.

Set A is proper subset of set B is denoted by "$A \subset B$"

In proper subset Equality Condⁿ is not allowed

Slide

Note:

1. if $A \subseteq B$ & $B \subseteq A$, then $A = B$ ✓
2. if $A \subseteq B$ & $B \nsubseteq A$, then $A \subset B$ ✓
3. Every set is a subset of itself.
   i.e. $A \subseteq A$ is always true.
4. ϕ is a subset of Every set.
   i.e. $\phi \subseteq A$ is always true w.r.t. any set A.

eg. Let $A = \{a, b, c\}$

Subsets of Set A are $\Rightarrow$ $\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}$

Proper subsets of Set A are $= \{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}$

All subsets are Proper subsets

Except Set itself.



2 mins Summary

Topic Types of sets

Topic Number of subsets, Concept of power set, and Cardinality of power set of a set

THANK - YOU

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